

Title

Shopz MERN Stack E-Commerce Web Application Professional Project Report This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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Executive Summary

Overview of the Shopz platform, objectives, technologies, and expected outcomes. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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Project Objectives

Build a scalable, secure, responsive e-commerce application using MERN. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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System Architecture

Frontend (React/Vite), Backend (Express), Database (MongoDB), JWT auth, REST APIs. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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Folder Structure

Detailed backend, frontend, docs, configuration, middleware, controllers, models, routes. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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Backend Design

Authentication, products, orders, middleware, validation, error handling, seeding. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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Frontend Design

Responsive UI, Context API, routing, reusable components, protected pages. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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Database Schema

Users, Products, Orders, Reviews, relationships and indexing strategy. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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REST API

Authentication, products, cart, checkout, orders, admin endpoints with payload concepts. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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Security

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UI/UX

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Testing

Manual testing, build verification, seed scripts, integration scenarios. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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Deployment

Docker, environment setup, production build, reverse proxy recommendations. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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Future Enhancements

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Conclusion

Summary of benefits and production readiness recommendations. This section elaborates architecture, implementation choices, best practices, diagrams (to be inserted), workflows, and practical considerations for development, deployment, scalability, maintainability, and user experience.

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